ROCSCIENCE

Case2_Search_Iter_1
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:45:25 am
Software Version: 9.041

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Slide2 Analysis Information

Case2_Search_Iter_1

Project Summary

File Name:	Case2_Search_Iter_1.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.255s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:45:25 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	7
Tolerance:	0.001
Maximum number of iterations:	50
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	1
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	20 kPa
Phi	20 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	2.113300
Center:	15.020, 24.465
Radius:	15.231
Left Slip Surface Endpoint:	10.100, 10.050
Right Slip Surface Endpoint:	28.565, 17.500
Resisting Moment:	13480.9 kN-m
Driving Moment:	6379.05 kN-m
Total Slice Area:	62.4888 m2
Surface Horizontal Width:	18.4651 m
Surface Average Height:	3.38416 m

Method: spencer

FS	2.107810
Center:	15.020, 24.465
Radius:	15.231
Left Slip Surface Endpoint:	10.100, 10.050
Right Slip Surface Endpoint:	28.565, 17.500
Resisting Moment:	13445.9 kN-m
Driving Moment:	6379.05 kN-m
Resisting Horizontal Force:	775.338 kN
Driving Horizontal Force:	367.84 kN
Total Slice Area:	62.4888 m2
Surface Horizontal Width:	18.4651 m
Surface Average Height:	3.38416 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	1562
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	1534
Number of Invalid Surfaces:	28

Error Codes
Error Code -108 reported for 17 surfaces
Error Code -111 reported for 11 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.1133

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.63787	51.7933	-13.7302	Material 1	20	20	13.4096	28.3385	22.91	0	22.91	19.6336	19.6336
2	2.63787	142.803	-3.63815	Material 1	20	20	18.9955	40.1432	55.3432	0	55.3432	54.1354	54.1354
3	2.63787	209.082	6.33802	Material 1	20	20	22.6812	47.9322	76.743	0	76.743	79.2623	79.2623
4	2.63787	250.3	16.5186	Material 1	20	20	24.5525	51.8867	87.608	0	87.608	94.8894	94.8894
5	2.63787	263.344	27.2926	Material 1	20	20	24.4828	51.7394	87.2034	0	87.2034	99.8359	99.8359
6	2.63787	235.75	39.3106	Material 1	20	20	21.7849	46.0381	71.539	0	71.539	89.3764	89.3764
7	2.63787	96.7029	54.2627	Material 1	20	20	12.7313	26.905	18.9714	0	18.9714	36.6645	36.6645

Global Minimum Query (spencer) - Safety Factor: 2.10781

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.63787	51.7933	-13.7302	Material 1	20	20	14.7443	31.0781	30.4367	0	30.4367	26.8342	26.8342
2	2.63787	142.803	-3.63815	Material 1	20	20	20.4321	43.067	63.3759	0	63.3759	62.0768	62.0768
3	2.63787	209.082	6.33802	Material 1	20	20	23.5364	49.6103	81.3537	0	81.3537	83.9679	83.9679
4	2.63787	250.3	16.5186	Material 1	20	20	24.5426	51.7311	87.1802	0	87.1802	94.4587	94.4587
5	2.63787	263.344	27.2926	Material 1	20	20	23.5858	49.7144	81.6396	0	81.6396	93.8093	93.8093
6	2.63787	235.75	39.3106	Material 1	20	20	20.311	42.8118	62.6747	0	62.6747	79.3054	79.3054
7	2.63787	96.7029	54.2627	Material 1	20	20	12.294	25.9135	16.2471	0	16.2471	33.3325	33.3325

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.1133

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	10.1001	10.05	0	0	0
2	12.7379	9.40552	50.1286	0	0
3	15.3758	9.2378	109.504	0	0
4	18.0137	9.53079	146.832	0	0
5	20.6516	10.3131	143.044	0	0
6	23.2894	11.6742	88.9176	0	0
7	25.9273	13.8341	-8.14891	0	0
8	28.5652	17.5	0	0	0

Global Minimum Query (spencer) - Safety Factor: 2.10781

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	10.1001	10.05	0	0	0
2	12.7379	9.40552	58.5079	18.9979	17.989
3	15.3758	9.2378	123.031	39.9492	17.989
4	18.0137	9.53079	161.277	52.3679	17.989
5	20.6516	10.3131	157.812	51.2427	17.989
6	23.2894	11.6742	108.907	35.3627	17.9889
7	25.9273	13.8341	27.1108	8.80307	17.989
8	28.5652	17.5	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 0 35, 0 35, 17.5 25, 17.5 10, 10 0, 10